BIO-RAD

Gene Pulser® Electroprotocols

*** We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.**

Cell Type Mammalian, adherent

Species Used Human, HeLa, epithelial carcinoma

Molecules Electroporated DNA: supercoiled DNA used for transient transfections; linearized DNA used for stable transfections.

Before the Pulse

Cell Growth Medium DMEM, 10% Fetal Calf Serum [FCS] (GIBCO/BRL, Sigma)

Growth Phase at Harvest 50 to 70% confluency

Pre-pulse Incubation 4° C, 10 min.

Wash Solution Wash two times in electroporation buffer

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature Room temperature

Electroporation Medium* HEPES Buffered Saline, 6mM glucose

Cell Density 5 x 10(6) cells/pulse for transient assay;

Volume of Cells 0.5 ml

DNA Concentration 10 µg / pulse

DNA Resuspension Buffer Not given; pulse volume: 0.8 ml

Volume of DNA Not given; pulse volume: 0.8 ml

Cuvette Gap 0.4 cm

Voltage 0.170 kV

Field Strength 0.425 kV/cm

Capacitor 960 µF

Resistor (Pulse Controller) Ω none

Time Constant 30 msec

After the Pulse

Outgrowth Medium DMEM, 10% Fetal Calf Serum (FCS)

Outgrowth Temperature 37 °C

Length of Incubation 48 to 72 hrs.

Selection Method or Assay Used G418 (stable transfections) and transient assays

Electroporation Efficiency Not given

Per Cent Survival about 50%

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Note: Stable transfections generally do not use carrier DNA. Also, the level of selective agent required to kill off non-transfected cellsneeds to be established before transfection. The level required should kill non-transfected cells in approximately 7 days.

HBS: 10mM HEPES,pH 7.2,150 mM NaCl, 5 mM CaCl2

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